

QUESTION 1

Create an array A with elements (12, 13, 14, 15, 16) and display them.

```
A <- c(12, 13, 14, 15, 16)
A
```

OUTPUT

```
[1] 12 13 14 15 16
```

QUESTION 2

Find the sum of all elements of A

```
sum(A)
```

OUTPUT

```
[1] 70
```

QUESTION 3

Find the product of all elements of A

```
prod(A)
```

OUTPUT

```
[1] 524160
```

QUESTION 4

Find the maximum and minimum element of A

```
max(A)
```

```
min(A)
```

OUTPUT

```
[1] 16 [1] 12
```

QUESTION 5

Find the range of array A

```
range(A)
```

OUTPUT

```
[1] 12 16
```

QUESTION 6

Find the mean, variance and standard deviation and median of values of A

```
mean(A)
var(A)
sd(A)
```

OUTPUT

```
[1] 14[1] 2.5[1] 1.581139
```

QUESTION 7

Sort the elements of A in both increasing and decreasing order and store them in B and C

```
B <- sort(A)
B
```

```
C <- sort(A, decreasing=TRUE)
C
```

OUTPUT

```
[1] 12 13 14 15 16[1] 16 15 14 13 12
```

QUESTION 8

Create a matrix of 3x4 to have the set of natural numbers

```
mat <- matrix(seq(1, 12), nrow=3, ncol=4)
print(mat)
```

OUTPUT

```
      [,1] [,2] [,3] [,4]
[1,]     1     4     7    10
[2,]     2     5     8     9
[3,]     3     6     9    12
```

QUESTION 9

Create MxN matrix by combining A, B and C row-wise (RW) and column-wise (CW).

```
rw <- rbind(A, B, C)
print(rw)
```

```
cw <- cbind(A, B, C)
print(cw)
```

OUTPUT

```
      [,1] [,2] [,3] [,4] [,5]
A      12   13   14   15   16
B      12   13   14   15   16
C      16   15   14   13   12
      A  B  C
[1,] 12 12 16
[2,] 13 13 15
[3,] 14 14 14
[4,] 15 15 13
[5,] 16 16 12
```

QUESTION 10

Find the 10 and 3 row elements of RW

```
print(rw[10])  
print(rw[3])
```

OUTPUT

```
[1] 15  
[1] 16
```

QUESTION 11

Find 1 and 4 columns of CW

```
print(cw[1])  
print(cw[4])
```

OUTPUT

```
[1] 12  
[1] 15
```

QUESTION 12

Using both RW and CW find sub-matrices having elements [2, 3] and [2, 4]

```
print("First Matrix")  
print(rw[2, 3])
```

```
print("Second Matrix")  
print(rw[2, 4])
```

OUTPUT

```
[1] "First Matrix"  
B  
14  
[1] "Second Matrix"  
B  
15
```

```
print("First Matrix")  
print(cw[2, 3])
```

```
print("Second Matrix")  
print(cw[2, 4])
```

OUTPUT

```
[1] "First Matrix"  
C  
15  
[1] "Second Matrix"
```